Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/30/2007
PI Name:	Sangani, Ashok S. Ph.D.		
Project Title:	Behavior of rapidly sheared bubble suspensions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	13244-1190	**Congressional District:**	25
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	09/01/1996	**End Date:**	08/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Koch, Donald (Cornell University)		
Grant/Contract No.:	NAG3-2427		
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Performance Goal Text:			
Task Description:	The objective of this experiment is to test the equations of motion of bubble suspensions in an ideal setting where potential flow prevails, and thereby validate the concepts of bubble pressure and effective viscosity of bubble suspensions. The conditions that promote potential flow are that the Reynolds number be large and the Weber number be small. The means to achieve the objectives are to generate a bubble suspension in a Couette cell in the microgravity environment and to shear it by rotating the outer cylinder.The forces acting on the bubbles in this configuration are the centrifugal force which pushes the bubbles toward the inner cylinder and the shear induced bubble phase pressure which tends to push the bubbles apart from each other. To accomplish the objective, measurements of the volume fraction, and the liquid and bubble velocity will be taken as functions of radial position. In addition the shear stress at the wall will be measured. Ground-based research objectives are: (i) to produce nearly monodisperse suspensions of nearly spherical,		

Task Description:	non-coalescing bubbles; (ii) to develop probes for measuring volume fraction and velocity distributions of the bubbles as a function of position; (iii) to measure wall shear stress; (iv) to perform experiments for bubble suspensions in vertical and inclined channels and in circular, vertical pipe configurations; and (v) to continue analytical work related to the description of flows of bubble suspensions. Multphase flows occur in many industrial processes. The questions concerning the stability of well-dispersed multiphase medium and its properties are important in predicting the behavior of multiphase flows and in designing equipment in which these flows occur. The detailed investigation of the idealized bubble suspensions under consideration is a first step toward developing an analytical framework for answering these questions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bubbly liquids are encountered in nature and in many industrial processes. These processes can be designed better if the properties of bubbly liquids and their equations of motion are known. Understanding forces responsible for seggregation and clustering and coalescence is crucial for designing reliable equipment, e.g. bubble reactors and gas-liquid transport lines, that handle gas-liquid flows.
Task Progress:	To understand how bubble-wall interactions may affect the flow behavior of bubble suspensions, and to determine the effective boundary conditions for the bubble-phase continuum, we analyzed in detail a model system which incorporates observed behavior of bubble-wall interactions as reported by Tsao and Koch (1997), Zenit et al (2001), and Tsang (2005). These investigators found that the component of the bubble velocity parallel to the wall reduces significantly due to its encounter with the wall while the normal velocity component may reverse with relatively small loss. Accordingly, we assume that the velocity of the bubble after “collision” with the wall is times the velocity before the collision. We also assume that the kinetic energy associated with the bubble motion is times the energy before the collision. In experiments, both and vary somewhat with bubble radius and the angle between the incident velocity of the bubble and the wall. In our model system, however, these parameters are treated as constants. Since our primary interest was in understanding how such bubble-wall collisions affect the behavior of bulk suspension, our model neglected hydrodynamic interactions among bubbles and the lift force due to variation of the velocity of the liquid. The bubbles were modeled as rigid particles with mass equal to the virtual mass of a spherical bubble. The collision between any two bubbles in the suspension was assumed to be perfectly elastic while the collision of a bubble with the wall was assumed to be described by the aforementioned parameters alpha and gamma. Finally, it was assumed that each bubble is acted upon by a buoyancy force and a viscous drag force as given by the well-known Levich expression. Molecular dynamics-like numerical simulations of this model system were carried out for various values of overall bubble volume fraction phi, ratio of (vertical) channel width to bubble radius, Reynolds number based on the terminal velocity of bubbles and their radius, and the parameters alpha and gamma. Profiles of bubble velocity, volume fraction, bubble-phase temperature, defined as one-third the bubble velocity variance, and other bubble-phase related properties were determined as a function of x2, distance from the center of the channel. To see how these profiles compare with the predictions based on the continuum-based averaged equations given by Spelt and Sangani (1997), we must specify the boundary conditions for the bubble phase. One of the obvious boundary conditions is that the velocity component of the bubble phase normal to the wall must vanish at the channel walls. The velocity component parallel to the wall, however, is nonzero. A slip boundary condition was derived by estimating the flux of the momentum lost by the bubble phase upon collisions with the wall and equating it to the bubble-phase shear stress. The velocity distribution of the bubbles was assumed to be isotropic Maxwellian for the purpose of estimating the momentum flux. Similarly, the energy lost upon collision was estimated and equated to the flux of fluctuation energy to provide a boundary condition for the bubble-phase temperature equation. We compared the results of simulations with the theory based on averaged equations and the above effective boundary conditions thus derived. It was found that the agreement is reasonable at moderate volume fractions (about 0.1 to 0.15). However, considerable deviations occur at lower volume fractions. On closer inspection, it was found that the velocity distribution of bubbles near the wall is considerably anisotropic and bimodal. Thus, boundary conditions listed above and the averaged equations of bubbly liquid are no longer applicable in the wall region. We developed a more elaborate kinetic theory of bubble suspensions to account for these effects. Briefly, the bubble-phase was treated as consisting of a mixture of two species: bubbles whose last collision was with a wall (wall-excited bubbles) and bubbles whose last collision was with another bubble (normal bubbles). Thus, a collision between a wall-excited bubble and a normal bubble leads to two normal bubbles while a collision of a normal bubble with a wall produces a wall-excited bubble. The velocity distribution of each species was assumed to be given by an anisotropic Maxwellian for each species. We developed a method for determining the collision integrals that appear in the kinetic theory of bidisperse systems for the collision contribution to stress and the source terms for anisotropic Maxwellian velocity distributions. The eight-fold integrations required are carried out by a combination of analytical and numerical methods. The equations we derived for bidisperse bubble-phase can also be used with minor modifications to granular/particulate flows. The analogous equations and closure relations for bidisperse granular systems have been proposed by Jenkins and Mancini (1987). We have shown that the simple relations proposed by Jenkins and Mancini do not provide accurate estimates of the stress tensor since the velocity distribution is considerably anisotropic. The method and detailed comparison will be described in Ozarkar et al. (2007). In summary, bubble-wall collisions significantly alter the velocity distribution of the bubbles near the walls so that the standard kinetic theory based on nearly isotropic Maxwellian velocity distribution does not predict the behavior bubble suspensions accurately near the walls. A kinetic theory that accounts for anisotropy and bimodal nature of velocity distribution was developed and shown to yield excellent predictions. REFERENCES Jenkins, J. T. and Mancini, F. 1987 Balance laws and constitutive relations for plane flows of a dense, binary mixture of smooth, nearly elastic, circular disks. J. Appl. Mech. 54, 27. Ozarkar, S. S., Kushch, V. I., Sangani, A. S., and Koch D. L. 2007 A kinetic theory for the effect of bubble-wall

interactions on the properties of bubble suspensions in a vertical channel (in preparation).

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Zenit, R., Koch, D.L., and Sangani, A.S. 2001 Measurements of the average properties of a suspension of bubbles rising in a vertical channel. J. Fluid Mech. 429, 307.

Bibliography Type:	Description: (Last Updated: 04/02/2007)
Articles in Peer-reviewed Journals	Ozarkar SS, Kushch VI, Sangani AS, Koch DL. "A kinetic theory for the effect of bubble-wall interactions on the properties of bubble suspensions in a vertical channel." Submitted to Physics of Fluids, March 2007. , Mar-2007
Books/Book Chapters	Sangani AS, Ozarkar SS, Tsang YH, Koch DL. "Flow of bubbly liquids in a vertical pipe: Theory and experiments." in "Fluid Mechanics and Its Applications. Proceedings of an IUTAM Symposium held at Argonne National Laboratory, October 4–7, 2004." Ed. S. Balachandar and A. Prosperetti. Dordrecht, The Netherlands : Springer, 2006, p. 381-392. http://dx.doi.org/10.1007/1-4020-4977-3_37 , Jan-2006
Dissertations and Theses	Tsang YH. "Experimental studies of millimeter-sized bubbles in electrolytic solutions --- rise velocity, coalescence, and vertical pipe flow. " Ph.D. Thesis, Cornell University, Ithaca, NY, 2005. , Dec-2005